

AQI-MLOps

Team 1

Data Science Capstone Project
Data Acquisition and Pre-Processing Report

Date:

05/04/2026

Team Members:

Name: Daniel

Name:

Name:

Name:

[The purpose of this report is to describe the data of your project. It includes three major sections: Data Sources, Data-Processing, and Appendix]

Identifying Data

Data Sources:

[Identify the data sources of your project. It may have more than one data source. Describe each of them and explain why you select the data sources.]

The Project relies on OpenWeatherMap (<https://openweathermap.org/>) they have both hourly and batch . We collect a bulk batch for 1 year of data source for 99 cities and create a hourly batch ingestion for each cities 1 hour data using lat , long to identify for the cities

```
Pipeline date range : 2025-03-20 → 2026-03-20
Chunk size           : 7 days per request
Request sleep        : 0.5 s
S3 bucket            : weather-bulk
Total cities         : 99
```

city	lat	lon
Tokyo	35.6762	139.6503
Delhi	28.6139	77.2090
Shanghai	31.2304	121.4737
Sao Paulo	-23.5505	-46.6333
Mexico City	19.4326	-99.1332
Cairo	30.0444	31.2357
Mumbai	19.0760	72.8777
Beijing	39.9042	116.4074
Dhaka	23.8103	90.4125
Osaka	34.6937	135.5023

Here is the data schema of the data.

Column	Type	Source	Description
timestamp	TIMESTAMP (UTC)	Both	Observation time
city / city_slug	STRING	Config	City name & URL slug
aqi	INT (1–5)	AQI API	Overall air quality index
co	DOUBLE (µg/m³)	AQI API	Carbon monoxide
no	DOUBLE (µg/m³)	AQI API	Nitric oxide
no2	DOUBLE (µg/m³)	AQI API	Nitrogen dioxide
o3	DOUBLE (µg/m³)	AQI API	Ozone
so2	DOUBLE (µg/m³)	AQI API	Sulphur dioxide
pm2_5	DOUBLE (µg/m³)	AQI API	Fine particulate matter
pm10	DOUBLE (µg/m³)	AQI API	Coarse particulate matter
nh3	DOUBLE (µg/m³)	AQI API	Ammonia
temp_c, humidity_pct, ...	DOUBLE	Weather API	Meteorological features (supplemental)
source	STRING	Pipeline	'pipeline' or 'live'

Acquisition Process:

[Describe the data acquisition process. Is the dataset ready for download? How do you download the data? Do you need to write your own code to acquire the data from a public or private source? Describe how you do it. Are there multiple data sources? How do you integrate the data from multiple sources? Any other process involved in the acquisition process?]

Batch Backfill :

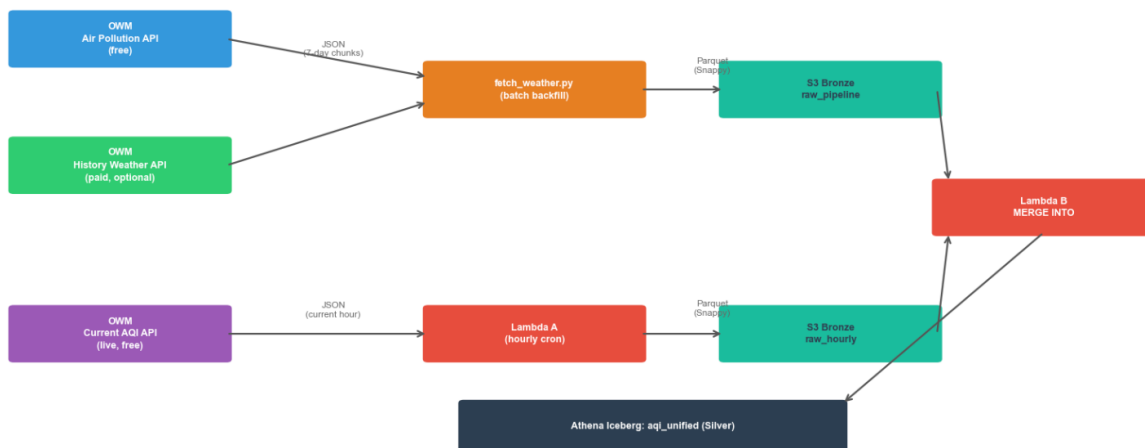
Written in Python (`fetch_weather.py`), config-driven via `config.yaml`. then splits the date range into 7-day chunks per city to avoid OWM API page limits. With a 0.5-second sleep between every request limits throughput to ≈ 2 req/s, well within OWM free-tier rate limits. This is bulk batch where we ingest 1 years of data first.

Live Hourly Lambda:

An EventBridge fires every hour. Lambda A calls the OWM Current Air Pollution API for each city, writes a Parquet file to `s3://weather-bulk/hourly/year=.../month=.../day=.../`, then notifies Lambda B. It runs `MERGE INTO aqi_unified` in Athena (ACID-safe deduplication on `timestamp + city_slug`), ensuring no duplicate rows from overlapping pipeline runs.

Data Integration:

Both pipelines land in separate bronze tables (`raw_pipeline`, `raw_hourly`). Athena `MERGE INTO` unifies them into the silver Iceberg table `aqi_unified`, tagged with a `source` column (`'pipeline'` or `'live'`) for lineage tracking.



Issues:

- Rate limitng: Long backfill times for 100 cities $\times$ 52 weeks , 7-day chunking + 0.5 s inter-request sleep keeps throughput at ≤ 2 req/s
- AQI class 6 present in some historical records (OWM sometimes returns 6 for extreme events, not in their documented scale) this is an unexpected label that XGBoost cannot handle so we convert 6 class to 5 during data loading

Data-Processing

[Examine the data you have acquired and understand the data properties. Is there any pre-processing you need to do before you can start analyzing the data? For example, missing data, sparsity, noise, veracity, ambiguity, interoperability, etc. Describe each data issue in a sub-section and explain how you clean up the data.]

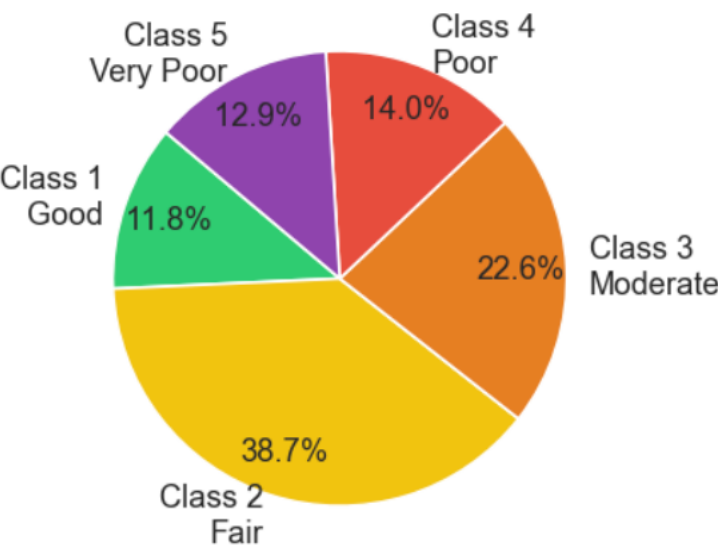
Missing data :

- Leading rows per city this is lag features require 3+ prior hours (the first 3 rows of each city's history have NaN lags).
- City day-boundaries , if a city is missing one hour's data, the lag chain breaks.

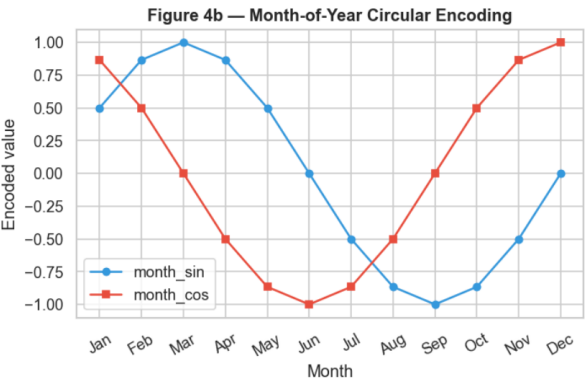
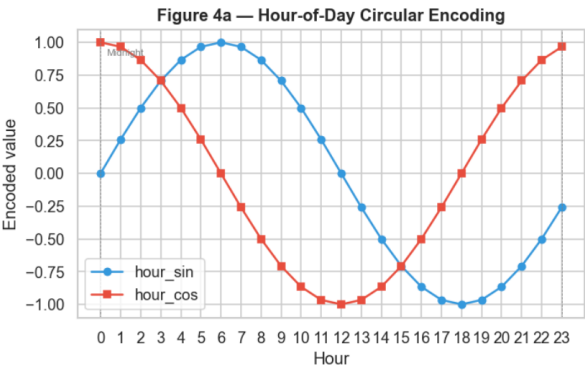
Strategies ; Drop rows where any required lag ('pm10_lag1', 'aqi_lag3', 'aqi_roll_std4') or the target ('aqi_next') is NaN — these are inherently untrainable.

Rows before feature engineering : 200
 Rows after feature engineering : 186 (dropped 14 rows due to NaN lags)
 Features available : 19

	pm10_lag1	aqi_delta_1h	pm10_delta_1h	aqi_delta_2h	aqi_delta_3h	aqi_roll_std4	pm25_delta_1h	hour_sin
count	186.000	186.000	177.000	186.000	186.000	186.000	167.000	186.000
mean	43.315	-0.022	-0.286	-0.011	-0.022	1.123	0.159	0.019
std	21.937	1.718	32.128	1.677	1.712	0.419	20.551	0.705
min	5.370	-4.000	-72.013	-4.000	-4.000	0.000	-40.520	-1.000
25%	22.995	-1.000	-21.627	-1.000	-1.000	0.816	-14.912	-0.707
50%	43.641	0.000	0.645	0.000	0.000	1.155	-0.433	0.000
75%	60.345	1.000	22.447	1.000	1.000	1.414	15.156	0.707
max	79.956	4.000	71.015	4.000	4.000	2.062	42.639	1.000



- As we see the class 4,5 are least dominate in the chart mostly class 2 and class 3.
- Raw hour (0-23) and month (1-12) are cyclical integers that break at boundaries (hour 23 → 0 are adjacent but numerically distant). We encode them as sine/cosine pairs so the model sees the correct circular distance.



- AQI is strongly auto-correlated the next hour's air quality depends heavily on recent history. We create a lagging feature help the training data

Feature	Formula	Semantics
pm10_lag1	PM10 at T-1	Historical particulate level
aqi_delta_1h	$AQI(T) - AQI(T-1)$	Rising (+) or falling (-)
aqi_delta_2h	$AQI(T) - AQI(T-2)$	Medium-term momentum
aqi_delta_3h	$AQI(T) - AQI(T-3)$	Sustained trend
pm10_delta_1h	$PM10(T) - PM10(T-1)$	Particulate momentum
pm25_delta_1h	$PM2.5(T) - PM2.5(T-1)$	Fine particulate momentum
aqi_roll_std4	$\sigma(AQI\ T-3..T)$	4-hour volatility

After dropping NA we use median imputate for training set avoid skew

```

feature
pm10_lag1          14.6800
aqi_delta_1h       0.0000
pm10_delta_1h      0.0500
aqi_delta_2h       0.0000
aqi_delta_3h       0.0000
aqi_roll_std4      0.0000
pm25_delta_1h      0.0300
hour_sin           0.0000
hour_cos           0.0000
month_sin          0.0000
month_cos          0.0000
pm25_ratio         0.1049
co                 156.1500
no                 0.0100
no2                2.7100
o3                 57.5500
so2                1.4800
nh3                0.5800
pm10               14.6800

```

This is the sample of evaluation after training model

Selected model : XGBoost Random Search
CV F1 (macro) : 0.9425
Val F1 (macro) : 0.9535

Best hyperparameters:

colsample_bytree	0.5996941730321137
gamma	0.15164172629634592
learning_rate	0.08095643339798969
max_delta_step	1
max_depth	6
min_child_weight	8
n_estimators	542
reg_alpha	0.9359393927098343
reg_lambda	6.449595315799922
subsample	0.659890472666743

Appendix

[Provide the code or pseudo code, data definition, sample data, and any other information in the appendix here.]

github repos : <https://github.com/danhdanhtuan0308>

| `data-pipeline/config.yaml` | Cities list, date range, S3 config, API URLs |

| `data-pipeline/fetch\_weather.py` | Batch backfill: calls OWM APIs, parses & uploads Parquet |

| `data-pipeline/lambda/handler.py` | Hourly live ingest: EventBridge → Lambda A |

| `lakehouse/merge\_handler.py` | Lambda B: Athena MERGE INTO for deduplication |

| `lakehouse/setup.sql` | Athena DDL: Bronze + Silver + Gold tables |

| `ml/train.py` | `engineer\_features()` + `build\_arrays()` |

| `ml/model-registry/median.json` | Persisted median values for inference imputation |

| `ml/model-registry/features.json` | Canonical 19-feature list used by training and serving |

Here is the example of inference : predict for a class

Sample processed feature row (first complete row from synthetic data):

pm10_lag1	8.8978
aqi_delta_1h	1.0000
pm10_delta_1h	36.6994
aqi_delta_2h	2.0000
aqi_delta_3h	0.0000
aqi_roll_std4	0.9574
pm25_delta_1h	2.1104
hour_sin	0.7071
hour_cos	0.7071
month_sin	0.0000
month_cos	-1.0000
pm25_ratio	0.3213
co	303.4283
no	3.1871
no2	24.7863
o3	22.0973
so2	5.8304
nh3	0.6825
pm10	45.5972
aqi_next	2.0000 ← TARGET

Table of Contributions

The table below identifies contributors to various sections of this document.

	Section	Writing	Editing
1	Data Sources	Daniel	Daniel
2	Data Pre-Processing	Daniel	Daniel
3	Appendix	Daniel	Daniel

Grading

The grade is given on the basis of quality, clarity, presentation, completeness, and writing of each section in the report. This is the grade of the group. Individual grades will be assigned at the end of the term when peer reviews are collected.